Panel Discussion

One particularly relevant question in politics today is by no means new. Political figures, philosophers, and other thinkers have inquired about the role of science in society throughout its history. This is due, in large part, to the way that science has been perceived by society, and the nature of scientific work itself. Investigating the history of science as an intellectual activity, as well as the arguments for and against its being supported, may offer an answer to this question.

Throughout its history, the distinctive and polarizing character of science brought many philosophical issues to center-stage in society. Science revolutionized hygiene and medicine, but also caused developments in warfare technology and industrialization. Figures such as Francis Bacon, Rene Descartes, and Jean-Jacques Rousseau would explore the conflicts driven by these issues, and offered commentary on how society can address them. These early commentaries ultimately led to government action through the foundation of learned societies and institutions dedicated to the sustainable and ethical development of science.

In my research, I have carefully investigated the perspectives of thinkers throughout the history of science, and the nature of the institutions dedicated to its advancement, in order to better understand the criticisms that the study has faced, and the ways that these criticisms have been addressed in theory or practice. From the Renaissance period onward, I have noted four characteristic features of the relationship between science and society, on the basis of which I will propose a model, describing the foundation of a scientifically-active, but fundamentally ethical society.

The first characteristic is science's apparent antagonism to morals. This idea is first addressed in the philosophical thought of Bacon and Descartes. Both claim in their writings that scientific activity is fully consistent with religion, and therefore morality. In the Enlightenment period to follow, Rousseau instead argued that this antagonism is naturally unavoidable, and can only be mitigated through the establishment of societies dedicated to the cultivation of both morals and science.

The second characteristic is the unpopularity of science, in other words, that there

are barriers to the public understanding and interest in the sciences. In economic terms, this prevents science from generating its own demand, and large-scale scientific projects such as those suggested by Francis Bacon become outright impossible. In order for such projects to be feasible, external forces on the market are necessary.

To explain why such forces may be desirable, the third characteristic is that the outputs of scientific activity appear to be useful. This utility takes three distinct forms: technological utility, civic utility, and national political utility. Technological utility is its potential to discover fundamental natural truths and put these to practical use. Civic utility finds its root in scientific education: even at a fundamental level, scientific education makes an educated people. Adam Smith suggests in his Wealth of Nations that this is key to the stability and security of free government. National political utility is more modern. Scientific institutions foster the development of science for national security and also promote international friendship.

Based on the insufficiency of natural demand, and the utility justifying market intervention, the fourth and final characteristic is that scientific activity requires external funding. Thinkers as early as Bacon and Descartes have affirmed that scientific activity requires leisure. This prevents scientists from sufficiently funding their own work. This funding can be either private or public; Bacon appealed to King James I in saying that scientific activity can only be supported "by public designation and expense," and Smith likewise says that the results of funding public education outweigh its costs. Some thinkers, such as Benjamin Franklin, instead argue that large-scale scientific and educational projects could be supported by private means, such as membership dues and philanthropy. Note, however, that both sources of funding can place limitations on scientific research. Public funding renders science subject to the volatility of the political realm, and private funding renders it subject to the interests of the funding agent.

These considerations frame the problem: science is valuable, but incapable of sustaining itself. The model I propose aims to address this problem. In this model, I assume that the government broadly works in the interests of the people or the regime,

so that government policy reflects public interest. I divide the model into two societal projects, the Cultural Project, and the Institutional Project. The Cultural Project is first, and is incited by science-minded individuals who are willing to initiate social change. It has two divisions: the first and most important of these is the institution of scientific education on a liberal arts basis. This has the dual function of motivating students to be interested in science by studying history, but also in improving the civic character of the society by studying philosophy. Despite its instrumental utility in inspiring the study of science, it is important that the liberal arts are studied for their own sake, so that its civic function is not lost. The other component of the Cultural Project is the publication of science, both to the scientific world and to the public at large. This mitigates mistrust towards the sciences. Eventually, the children educated in this Project will have grown up, and will ideally hold a general interest in the sciences. This interest will then drive the establishment of, and later, the investment in institutions dedicated to scientific development. This is the Institutional Project. The education of the public will prevent against many of the ills that come from public or private funding. These institutions will then sponsor the publication of science results, including the technological developments that come from these results. These factors then feed back into the education of young scientists. At this stage, the Cultural and Institutional Projects have a reciprocal relationship, each contributing to the other's development. So long as both of the projects are maintained, this results in a stable, scientifically minded, and fundamentally ethical society.

This model results from the writings of key thinkers in the history of philosophy and science. In synthesizing their conceptions of scientific development, I am hoping to offer a way for the supporter of science to address many of the public misconceptions and, indeed, many of the societal problems brought on by scientific development. In the model, the methods proposed by key thinkers offer hope not only to enrich scientific activity, but also civic activity, for the promotion of not only a science-minded people, but also a reasonable and moral people.

On the History of Public Support for the Sciences

C. Schmidt

1 Introduction

For much of Western intellectual history, what we would now consider the hard sciences were not distinct from philosophy, but were collectively known as *natural philosophy*. Many scholars whose dominant intellectual contributions were philosophical also contributed to the early development of natural philosophy - in fact, the modern scientific method itself emerged from philosophical explorations into the notions of truth, utility, and human psychology.

It was not until the Renaissance period that natural philosophy became distinct in character from other variants of philosophy. Scientific discoveries presented challenges to the authorities and the *status quo* - many pioneers of science therefore did not see the fruition of their ambitions during their lifetime. In the succeeding centuries, however, both the public and the state recognized the benefit of promoting experimental research; therefore private and state-run institutions would emerge throughout Europe and the Americas for the promotion and development of natural philosophy. By the 19th century, natural philosophy further distinguished itself, becoming *science*: a vocation, rather than simply an activity of leisure. With this distinctive character came profound philosophical implications: the elevation of the sciences in society lead to widespread industrialization, advances in medicine, and advances in warfare.

Thinkers have therefore been concerned with the *role* of science in society throughout its history. To what extent should the public support the sciences? To what extent should the state? Does scientific inquiry warrant such support? This report investigates various philosophical perspectives on science, in order to show that they converge on certain features that clarify its role in society. The content of these perspectives will be illustrated in the form of

a model, whose purpose is to clarify the manner in which science can act as a public good. This model involves, broadly, two societal projects: the *cultural project* and the *institutional project*. The cultural project consists first and foremost of scientific education on a *liberal arts basis*, for the sake of inspiring youth to engage in the sciences, and instructing them how to do so ethically and sustainably. The institutional project involves, first, the establishment of institutions designated for scientific advancement, and then the subsequent maintaining of these institutions. The ultimate goal of these institutions is broadly to publish scientific work; on the one hand, scientific work should be published within the scientific community, and on the other, it should be publicized for the public at large to engage with, both via commercial scientific products (hygiene, communication, etc.) and through popular science publications. These outputs feed back into the cultural project, such that the two projects are ultimately in coexistence within the society.

2 Historical Outline

The philosophical basis for this report begins with the late Renaissance period of Europe. It was during this time that Francis Bacon would frame his philosophical thought. Despite a lengthy political career, his philosophical output during this period was limited. His most notable work released during this time was his *On the Advancement of Learning* in 1605. In this work, Bacon argues against various moral and political objections to the advancement of the sciences. The dedication of the second part of this work, addressed to King James, is of particular interest due to its calling for direct government endowment of natural philosophy and highlighting the limitations private endowment imposes on scientific development. However, after 1620, Bacon had published most of his philosophical works, including a revised version of his 1605 piece. These later works are collectively known for framing a large-scale project called the *Great Instauration* - of which the *Advancement* was the first component. Following it in 1620 was the *New Organon* - a work which formulates a new method for

natural philosophy, now known as the *scientific method*. He frames this method on the basis of philosophical considerations, primarily the forces affecting the presentation and discovery of empirical truth, as described by his *Four Idols*. The key idea expressed through Bacon's *Great Instauration* is the notion that science requires institutional support in order for its true utility to be realized.

The result of such support is illustrated best in his utopian novel *New Atlantis*. Despite being posthumously released unfinished in 1627 (Bacon and Sargent 239), Bacon illustrates in the narrative a model of society organized on the basis of science, including the moral character of such a society. He provides descriptions of political and academic institutions, as well as accounts of conversations between the European travelers and the citizens of the utopia, *Bensalem*, aimed at showcasing the compatibility of morality - particularly *Christian* morality - with the development of natural philosophy. Bacon's characterization of Bensalem provides a model for the discussion of how morality is affected by the sciences - a topic discussed extensively by later thinkers, and a key concern in the politics of science.

Ultimately, little of Bacon's philosophical thought would be realized during his lifetime. However, his political thought would heavily influence the Enlightenment philosophy of the next century, and would directly lead to the establishment of learned institutions in Europe.

René Descartes would also write on the societal role of science. Apart from his intellectual contributions to philosophy, Descartes was profoundly influential in the field of mathematics, and is credited with the creation of analytic geometry (Descartes 12). Dissatisfaction with the institutionalized, Jesuit education of his youth inspired Descartes to search out a more solid epistemological basis for philosophy - one promising the same certainty as that found in mathematics.

This basis is described in his *Discourse on Method*, with a scope broad enough not only to lend philosophy a basis of certainty, but also to prove the very existence of God. This is his notorious claim *cogito ergo sum* - "I think, therefore I am". Under this epistemological framework, he describes the extent of the truths that can be derived from it. The

investigation of these truths can rightly be considered *scientific inquiry*. In the final part of this discourse (33–44), Descartes describes the value of such inquiry: he first expresses his frustrations with the hostile intellectual climate of his day[1]. He then contrasts this by describing a *collaborative* view of intellectual culture. Driven by the widespread publication of, and engagement with, scientific research, his account emphasizes the practical value of such a culture and the conditions requisite for its existence.

The era during and immediately following Bacon and Descartes' lives was known as the *Scientific Revolution* for significant contemporary discoveries in natural philosophy. It was during this period that, for example, classical mechanics would be revolutionized by the formulation of Newton's laws of motion. These developments would in turn build a modern understanding of planetary motion and chemistry (Spencer et al.). The recognized value of these developments eventually coincided with increased public support for experimental philosophy, in the form of learned societies. In 1662, the *Royal Society of London* was established - deriving many of its aims and its structure from the Baconian project (The Royal Society 3–10). Four years later, the *Academie des Sciences* would be established in Paris. Many other learned groups would emerge during this period, some being privately funded such as the *American Philosophical Society* in 1743, but many of which would receive government endowment or protections.

One such society was the *Academy of Dijon,* whose 1750 essay competition would bring notoriety to another political philosopher: Jean-Jacques Rousseau. In his *Discourse on the Sciences and the Arts*, he pushes back on the status of scientific study in society and investigates the extent and the cause of its negative effects on the morality and social norms (the *mores*) of its citizens. One important remedy to this, following the Baconian tradition, is the institution of societies coupling the cultivation of mores to the development of science. Much of the apparent corruption fostered by science is derived from its being disempowered by that society; such societies, to Rousseau, mitigate this corruption (Rousseau 1–21).

[1]The same climate that, in 1633, would put fellow natural philosopher Galileo Galilei on trial for his account of heliocentrism (Descartes 23)

Rousseau thus provides a voice for the main criticism against the promotion of the sciences, and is therefore important in how the model will be framed in Section 4 below.

By the late 18th century, the empirical school of Bacon and his successors had given rise to utilitarian ethics and enhanced the study of economics. These systems, which rely heavily on considerations of *utility*, would become popular in shaping ethical and political thought. As such, the promotion of scientific education and research was justified by thinkers according to its utility. This utility refers predominantly to public benefit - especially benefit in terms of *production* and *comfort*. Pragmatic considerations would be emphasized by Benjamin Franklin in his founding of the aforementioned American Philosophical Society (Franklin 295–297). Franklin would later formulate a system of learning which emphasized practical knowledge and the ethical value of science from an early age (323–344). Economist Adam Smith would similarly consider the value of education in his *Wealth of Nations* (Smith 758–788). Later, reforms to the existing learned societies would also appeal to utilitarian concerns ("Lord Derby on the Endowment of Scientific Research"). Ethical theorists would, in turn, acknowledge the value of scientific development to the future of society[2].

During the mid-19th century, many of the learned societies would begin to stagnate, particularly in France (Shinn 291–302), largely due to heightened centralized control and subsequent loss of enthusiasm from scientists. Scientific developments globally would be less concerned with the notion of intrinsic value in scientific knowledge, but would not fall much into decline. Rather, the rise of industrialization would give the public unprecedented engagement with scientific developments, both in terms of labor requirements and the diffusion of consumer goods. Likewise, developments in war technology would dominate scientific research. By the turn of the century, the Gatling gun, the motorized submarine, and chemical warfare had been developed as a result of scientific activity. These developments signaled a shift in the public role of science: in terms of industrialization and the implications of scientific development on national defense, the public quickly became *dependent* on science

[2]See Bentham and Mill 105–106.

to an unprecedented extent.

These developments in war technology would characterize the First and Second World Wars. Scientific discoveries would be promoted on the public stage largely because of their implications on warfare. For example, the 1944 Nobel Prize in chemistry was awarded to Otto Hahn for his discovery of nuclear fission (Nobel Prize in Chemistry 1944). Around the same time, the *Manhattan Project* researched nuclear fission to design atomic weaponry, and was fully funded by the United States government and developed by the U.S. Army Corps of Engineers (Britannica Editors). The important role of the sciences in the Allied victory in the Second World War would inspire the foundation of new scientific organizations. The *National Science Foundation* (NSF) would be established by Congress in 1950. The *National Aeronautics and Space Administration* (NASA) was similarly formed eight years later.

3 Defining Features of the Model from Literature

The various thinkers' perspectives on the status of the sciences in society (and, more particularly, the degree to which they should be publicly supported) are nuanced; however, there are notable points at which their perspectives seemingly converge. I wish to highlight here four key features of the interaction between society and hard science as framed by the various thinkers listed in the previous section.

I: The sciences appear antagonistic to morals.

The notion that the sciences are antagonistic to morality pervades the writings of the early proponents of the sciences. This notion is driven by two sources of antagonism: first, that the development of the sciences is subversive to the authorities in existence, and second, that it poses a threat to the moral character of society at large.

In *The Advancement of Learning*, Bacon addresses the antagonism of the authorities especially by addressing the religious concerns encompassing scientific inquiry, such as the idea

that "the aspiring to over much knowledge was the original temptation and sin whereupon ensued the fall of man" (Bacon and Sargent 6). Bacon appeals to Christian scripture to outline three limitations, such that scientific inquiry is not inherently sinful nor immoral:

> The first, *that we do not so place our felicity in knowledge, as we forget our morality.* The second, *that we make application of our knowledge to give ourselves repose and contentment, and not distaste or repining.* The third, *that we do not presume by the contemplation of nature to attain to the mysteries of God.* (8)

Religious considerations, such as this one by Bacon, were written largely in response to pushback from religious authorities. Therefore their function is to show that scientific study is *not* overtly subversive to religious authority, and is in fact compatible with such authority under certain conditions. Descartes likewise argued throughout his *Discourse on Method* that scientific reasoning is compatible with Christian religion (and by extension, Christian morals). Much of his perspective regarding scientific culture is a response to the existing religious authorities' failure to recognize this compatibility (Descartes 34).

Rousseau argues from the perspective of this second source of apparent antagonism. In his *Discourse on the Sciences and the Arts,* he highlights many of the vices emergent from scientific study. Rousseau attributes these vices largely to its requiring *luxury*:

> Luxury seldom thrives without the sciences and arts, and they never thrive without it. [...] Precisely what, then, is at issue in this question of luxury? To know whether it is more important for empires to be brilliant and fleeting, or virtuous and long-lasting. I say brilliant, but by what luster? The taste for ostentation is hardly ever combined in the same souls with the taste for honesty. [...] In this way the dissolution of mores, a necessary consequence of luxury, leads in turn to the corruption of taste. (Rousseau 12–14)

A partial remedy to this is found in the establishment of societies dedicated to *both* the promotion of the sciences and the practice of ethics - referring to the many learned societies

of France and England that had emerged during the Scientific Revolution.

> [That great monarch] drew from the very bosom of the sciences and the arts, sources of a thousand disorders, those famed societies which are charged simultaneously with the dangerous trust of human knowledge and the sacred trust of mores, by the attention they pay to maintaining them in all their purity, and to requiring it in the members they admit. These wise institutions ... will at least serve as a restraint on men of letters... (17–18)

Rousseau therefore offers a more skeptical perspective towards the vocation of science. The "restraint" on intellectuals lies in their personal desire to take part in the learned institutions. This drives them to "try to make themselves worthy by means of useful works and irreproachable mores" (18). This can be contrasted to Bacon's perspective. He argues that certain internal constraints are sufficient to preserve the morality of scientific activity - these constraints being a proper understanding of its end, namely, the joining of action and contemplation, such that scientific activity has practical value (Bacon and Sargent 29). Rousseau, however, is more critical, characterizing the pursuit of scientific learning as unstable, and is only rendered a moral good under strict, *external* constraints.

II: The sciences are characteristically unpopular.

The claim that the sciences are characteristically unpopular can be better understood as a fundamental barrier between the interests of the public and those of scientists. Unlike the first point, this is not unpopularity in the way of *antagonism*, but rather just the public's not overtly embracing science. This arises in part from the technical nature of scientific research; the depth and scope of novel research presents difficulties in building public understanding, and fostering popular interest. This barrier also in part arises from the uncertain, non-linear progress of scientific research, such that maintaining popular interest and obtaining consistent material support is challenging for scientists. Understanding the implications of

public interest on the support of the sciences is therefore largely a question of *economics*. This section will largely rely on economic considerations to understand, and frame how thinkers have addressed, the unpopularity of the sciences.

Public interest in the sciences can rightly be considered *demand* in the economic sense, and the material endowment is essentially an *investment* in the sciences. Within the literature, such questions of demand and investment focus on scientific *education* which is instrumental in creating a culture literate and interested in scientific inquiry. This renders scientific education a mechanism, generating demand in the public for scientific research.

In Book V, Chapter 1 of *Wealth of Nations*, author Adam Smith explores the expenses of government, among these being the education of youth. Because the schools at the time required the scholar to pay his master a fee based on his performance, Smith considers the institutions of education a source of "natural revenue" (Smith 759). However, when such an institution is made public (therefore fixing the salaries of instructors), he claims that this has a detrimental effect on the quality of education received (760). The remainder of this section is dedicated to weighing the effects of this trade-off on the morality and politics of a nation.

Where education is a market good, he claims that it would depend on demand:

> Were there no publick institutions for education, no system, no science would be taught for which there was not some demand; or which circumstances of the times did not render it, either necessary, or convenient, or at least fashionable to learn. (780)

This has an effect on the propagation of scientific knowledge throughout a society. When taken to an extreme, such a system of education would highly depend on one's line of work; as such, many aspects of knowledge would be neglected for lack of demand. Scientific learning, as expressed by the institutions in e.g.: Bacon's *Bensalem*, require long-term intellectual and material investments; as such, their being subject to market demand limits the extent of their social utility to the most expedient demands of the day.

> Of the great and extensive interests of his country, [the laborer] is altogether incapable of judging; and unless very particular pains have been taken to render him otherwise, he is equally incapable of defending his country in war. [...] in every improved and civilized society this is the state into which the labouring poor, that is, the great body of the people, must necessarily fall, unless government takes some pains to prevent it. (782)

The reasons a government may have for preventing this will be discussed in III below.

The observation that science lacks public demand would be echoed around a century later by proponents of scientific funding. For example, by 1851, the Royal Society was granted a £1000 grant for the endowment of scientific research. However, public figures later called for increases in this endowment. In 1875, a speech by Edward Henry Stanley (Lord Derby) would be published in *Nature*, expressing optimism toward the yet-unrealized recognition of the sciences by wealthy philanthropists and the public at large ("Lord Derby on the Endowment of Scientific Research"):

> When have schools, hospitals, public parks, museums, institutes been more abundant than at the present day? Science has no endowments, or next to none; but only because the interest in that class of subjects is comparatively new, and rich men, who want to do some good with their capital, have not looked much in that direction as yet.

These perspectives have shown that the sciences are unpopular by construction: the technicality and intellectual investment required to advance science are not conducive to *naturally* producing public demand. External forces, such as education, philanthropy, and political action are therefore required to secure the development of the sciences. The justification for instituting such forces will likewise be discussed in III below.

III: However, the sciences are useful.

In the last section it has been shown that the sciences are insufficient to naturally produce public demand, and external political or economic forces are necessary for their development. However, many thinkers affirm that the utility of the sciences warrants the employment of these forces. In this section, three varieties of the utility of science will be explored: *technological* utility, *civic* utility, and *national political* utility.

The technological utility of scientific research refers to its potential to advance the discovery of fundamental truths, and to apply such discoveries for practical use. This form of utility is extensively described in Bacon's *New Organon.* In order to understand the manner and extent to which scientific research holds technological utility, Bacon distinguishes two types of experiment: *experimenta lucifera* and *experimenta fructifera* - experiments of *light* and *fruit* (Bacon and Sargent 130). Experiments of light are what might be considered today *basic research,* "not for the purpose of producing any particular effect, but only of discovering the natural cause of some effect". Experiments of fruit, by contrast, follow from experiments of light, and are indeed intended to produce results. Nevertheless, Bacon affirms that experimentation of both forms is useful:

> To suppose therefore that things like these are of no use is the same as to suppose that light is of no use, because it is not a thing solid and material. And the truth is that the knowledge of simple natures well examined and defined is as light. It gives entrance to all the secrets of nature's workshop, and virtually includes and draws after it whole bands and troops of works, and opens to us the sources of the noblest axioms and yet in itself is of no great use. (141)

This quote is intended to show that even basic research, which does not *directly* produce useful results, is useful in directing the course of *applied* research, which *may* produce technologically useful results. The ideal interplay between these two forms of research is illustrated by the institutions of Bensalem in his *New Atlantis.*

Descartes likewise recognizes the technological utility of the sciences in his *Discourse.* In his own search for basic physical truths through experimentation, Descartes notes the importance of *publishing* scientific results (Descartes 34–35). This is for two reasons: first, publishing science results helps to clear up important disagreements in fundamental scientific principles. Secondly, it allows the accumulation of knowledge over time - allowing the next generation of scientists to advance research beyond where their predecessors left off. The result of this is the possibility of a "practical philosophy" - geared towards the use of natural resources for our own *use*, rendering us "masters and possessors of nature" (35).

The civic utility of the sciences would be most prominently recognized by the utilitarians of the next century. This utility refers to the distinctly *moral* advantages which science, and particularly scientific education, renders to the public at large. This form of utility, as will be shown in the thinkers' perspectives below, is not entirely distinct from technological utility. Rather, its scope is more directly behavioral; as it induces the public to support science *for the sake* of its technological utility.

The pragmatic value of the sciences would inspire Benjamin Franklin to establish the American Philosophical Society in 1743 (Franklin 295–297). Like Descartes before him, Franklin acknowledged that the publication of science results (more precisely, in its initial form, agricultural results) had technological utility in its improving the economic status of the Colonies. However, in order to advance the sciences, their value must be recognized on a more fundamental and general level. This recognition throughout a society can be most directly fostered through the education of youth. Franklin would later write the framework for an educational system in his essay *Proposals Relating to the Education of Youth in Pensilvania*, which was designed to prepare students with technical knowledge in fields such as plant biology and mechanics for the service of mankind - and to generate interest in their understanding of such fields from an early age (323–343).

Smith also recognized the civic value of a scientifically-educated public in his *Wealth of Nations*:

> [An instructed and intelligent people] are more exposed to examine, and more capable of seeing through, the interested complaints of faction and sedition, and they are, upon that account, less apt to be misled into any wanton or unnecessary opposition to the measures of government. In free countries, where the safety of government depends very much upon the favourable judgment which the people may form of its conduct, it must surely be of the highest importance that they should not be disposed to judge rashly or capriciously concerning it. (Smith 788)

It is in the writings of Smith that the civic value of scientific learning[3] can be most directly understood. The prior thinkers offer accounts of the value directly produced by scientific activity, particularly from scientific research - namely, technological utility. Smith, however, offers a commentary on the behavioral effects of scientific education on the public. These behavioral effects define the civic utility of scientific learning: the reason exercised in obtaining a basic education (and, by extension, engaging in scientific activity) translates directly to social and political activity. As such, it prevents the nation from being susceptible to civil unrest.

The national political utility of the sciences is best expressed through the government institutions established to advance scientific research. This form of utility is broadly *strategic* in nature, as it incorporates multiple political (and military) interests: among these are the national prestige a country can gain by cultivating science, the fostering of international collaboration, as well as pure national defense. In order to showcase these features of the national political utility of science, the establishing acts for three government institutions will be discussed: the Royal Society, the NSF, and NASA.

In the 1663 charter of the Royal Society, founder King Charles II acknowledged the prestige gained through the promotion of scientific activity, and justified the establishment of the institution under the notion that "the whole world of letters may always recognize [the King]

[3]Note also that Smith, over and above Bacon, Descartes, and Franklin, is referring to science in the broad sense. This isn't exclusively what might be considered the hard sciences, but refers more to the kind of learning that emerges from a fundamental, public education. Note however that he does place particular value on the study of geometry and mechanics (Smith 785–786).

not only as the Defender of the Faith, but also as the universal lover and patron of every kind of truth" (The Royal Society 3). This is a reflection of the potential political affluence which the United Kingdom could gain from its promoting scientific activity. Broadly, the scope of the charter affirms the benefit of promoting the sciences to mankind, while also recognizing their political utility. It was also under this notion that public figures would call for improvements to the Royal Society, that is, increased endowments to scientific researchers, in the succeeding centuries ("Government Aid to Scientific Research").

The establishment of the NSF in 1950 was driven largely by a different motive - national defense. In the context of the rapid development of warfare technology in the late 19th century, ultimately leading up to the Allied victory in the Second World War, the United States government recognized the role of the *hard sciences* in developing the country's military power. The NSF was therefore designed exclusively to support the *hard sciences*. From Section 3 of the establishing act (U.S. Congress, National Science Foundation Act of 1950), its eight authorized functions can be broadly described as (1) promoting "basic research" in the sciences through political action and financial awards, and (2) evaluating the extent to which the promoted research is useful for national defense.

The 1958 establishment of NASA likewise was driven by a different purpose. In contrast to the clear and recognized role of national defense on the activity of the NSF, note the following clause from the National Aeronautics and Space Act of 1958:

> The Congress hereby declares that it is the policy of the United States that activities in space should be devoted to peaceful purposes for the benefit of all mankind. (National Aeronautics and Space Act of 1958)

More so than the establishing act of the NSF, NASA's establishment has an overarching moral objective: to materially promote *peaceful* research activity in space science. The clause shown establishes a limit on the purpose of the organization, despite the recognized implications that aeronautical activities hold on the "general welfare and security" of the United States. Nevertheless, as for the NSF, provisions are made for scientific discoveries

promoted by NASA to be applied to national defense.

A trend worth noting in the establishment of all of these institutions, however, is their consistent provision for collaborative scientific activity across national borders. This establishes another facet of the political utility of the sciences: scientific activity is an instrument of political diplomacy. The hallmark example of this is the engagement of NASA in the International Space Station (ISS). The ISS, according to the 2010 National Space Policy, is to be utilized for "for scientific, technological, commercial, diplomatic, and educational purposes" (Executive Office of the President of the United States 15) - illustrating the political and technological implications of the sciences in fostering peaceful international relations.

The utility of science is in its producing long-term, collective benefits for society: including technological advances, improvements to civic culture, and service as a political tool in improving national defense and fostering diplomacy. However, the very nature of these benefits prevents them from being adequately supported by the market alone - as stated in II - and imposes an important condition on the development of the sciences.

IV: The study of the sciences requires funding.

The previous three sections have shown that, in certain arrangements, the sciences are not morally subversive, but are nevertheless unable to advance themselves by appealing to public demand because their utility is long-term and collective. However, this utility has warranted the establishment of external forces, such as education and government institutions, in order to advance the sciences. The final point to emphasize is that these sciences must rely on external funding. The reasoning for this is twofold: first, the actual experimentation and scientific process is both expensive and not able to be self-funded, and second, the institutions established for scientific advancement themselves require funding to operate. Thinkers have assessed (and disagree on) the extent to which *private* funding is sufficient to advance research, or whether public funding is in fact necessary; the existence of state scientific institutions is an expression of the latter opinion.

It is a defining feature of scientific research that it requires leisure. This is stated explicitly in the works of Bacon and Descartes[4]. This is because of the significant time commitment which scientific research and experiment require. However, as stated, such experimentation is often expensive, and the time commitment required prevents scientists from being able to fund their own research. This is one of the defining arguments for the requirement of external funding for scientific research.

Bacon's dedication of the second part of his *Advancement of Learning* concludes with a statement to King James I affirming that his project for the development of the sciences was such that it could *not* be realized by private means, but must be done "by public designation and expense" (Bacon and Sargent 55). This is a consideration of *scope*. Bacon affirms that many scientific endeavors are outright impossible under certain circumstances - private funding alone being one of them.

Similarly, Smith considered the benefit of educational institutions (whether specifically scientific or not) vastly greater than the losses at the public expense (Smith 785). These perspectives together illustrate that, to the proponents of public funding, the benefits conferred by supporting scientific education and research compensate - or even outweigh - the costs involved in funding them.

Some thinkers, however, have affirmed that exclusively private funding for the sciences is sufficient to ensure their advancement. This perspective can be seen in Franklin's intellectual project in the Colonies. Though the prospects of Franklin's project were not as ambitious as Bacon's, the American Philosophical Society was indeed only reliant on membership dues for its operation. As such, only those who provided funding for the propagation of scientific knowledge and the endowment of funds to experimentation could directly receive its benefits (Franklin 297). Franklin would also deem the private contributions and service from "Persons of Leisure and publick Spirit" sufficient to establish his proposed academic system (325–326).

[4]See Bacon and Sargent 13, and Descartes 41.

In his speech advocating for science funding, Lord Derby recognized that both private and public sources of funding were at that point insufficient. He therefore showed no preference for either source of funding, but advocated for both. Nevertheless, a direct result of his efforts would be a fourfold increase in government endowment to scientific research granted to the Royal Society ("Government Aid to Scientific Research").

By the late 19th century, then, the Royal Society became an effective realization of the Baconian project through its dedication to directly funding scientific activity. With the United States' establishment of the NSF and NASA, scientific activity was fully institutionalized. Section 3a of the NSF's establishing act (U.S. Congress, National Science Foundation Act of 1950) explicitly authorizes the institution "to award... scholarships and graduate fellowships" to students of the hard sciences. NASA was likewise authorized in section 203a to "acquire, construct, improve, repair, operate, and maintain" the facilities required to carry out its scientific activities (National Aeronautics and Space Act of 1958). These three institutions together are the actual expression of the the thinkers' advocacy for public funding of the sciences.

Both sources of funding, however, place potential limitations on scientific development. Publicly-funded research is subject to volatile and short-term political interests, which has implications on the manifestation of some of its important forms of utility. However, private funding is often more limited in absolute quantity, and is subject to the desires of particular people or groups. This has two effects: first, it directly limits the scope of scientific activity, and second, it renders research subject to the particular whims of its funding sources. At best, this has the effect of simply directing privately funded research in accordance with the wishes of the funding source; at worst, it leaves scientific research susceptible to biases[5]. This understanding justifies the idea that the promotion of the sciences should be more fundamental than the institutions once established: it requires *ongoing cultural engagement* with the sciences, in order to direct both political and industrial activity in a way that advances the

[5]Industry sponsorship, particularly in the medical, alcohol, and tobacco industries, often introduces biases to align scientific findings with scholarly research. See Fabbri et al. e13-e14.

discovery of truth. Though public interest, as stated, is insufficient to support the sciences alone, it is an important force in directing scientific activity.

4 The Model

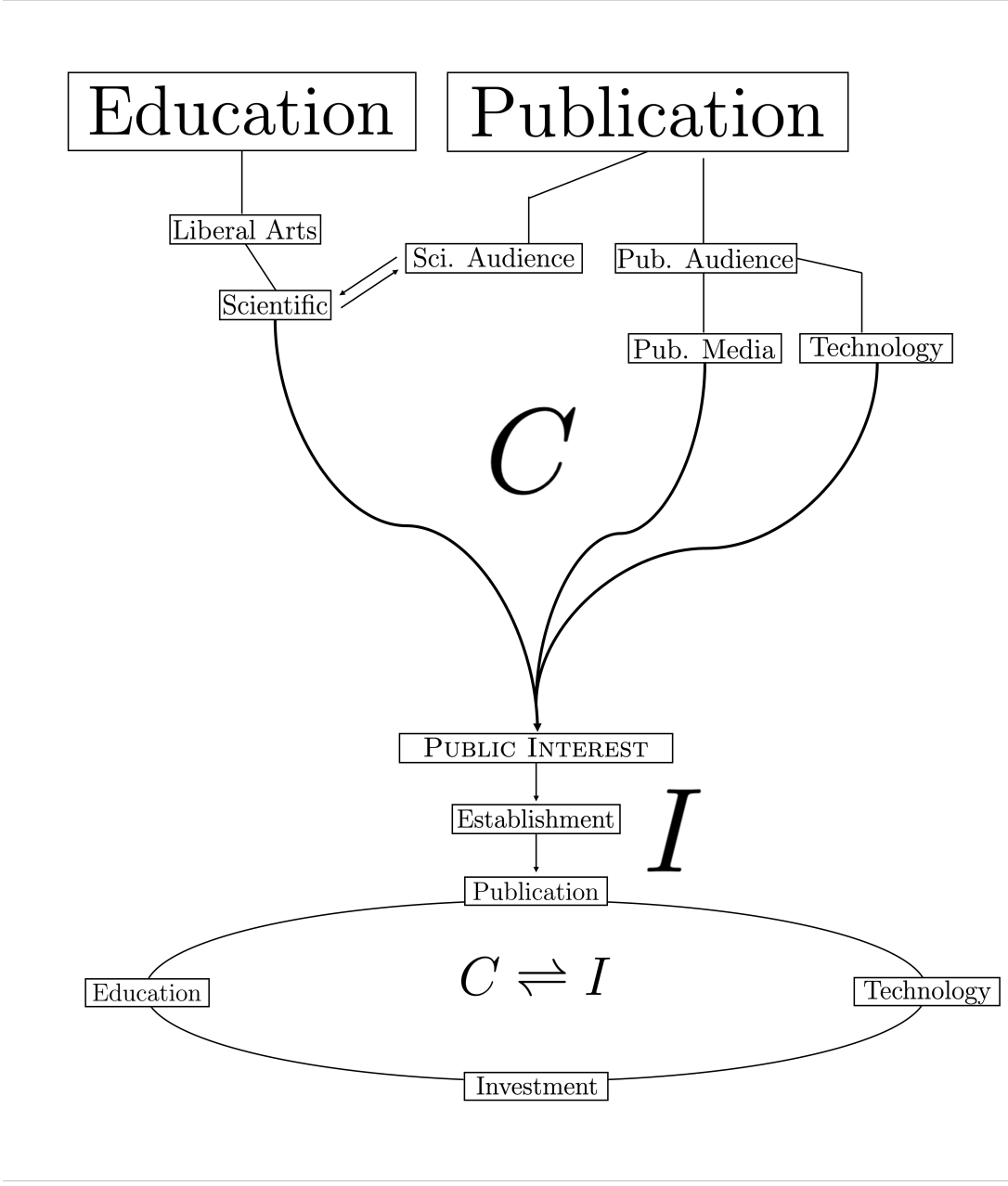


Figure 1: Schematic of the model.

As stated throughout Section 3, thinkers have attempted to address (I) the tension between morals and science, (II) the lack of public demand it generates, (III) that its potential long-term utility is not reflected by this demand, but requires external forces to be realized, and that (IV) these external forces, along with scientific activity itself, require external funding. In this section, I am putting forward a model, framed around these considerations and the perspectives of the various thinkers, in order to organically resolve these tensions. This model can be broadly divided into two parts: the *Cultural Project* and the *Institutional Project.* For the sake of simplicity, I am assuming that the government over the state works in the interest of its citizens, or at least in the interest of preserving the regime, so that the political actions taken by the government are broadly reflective of the public interest. In this state, there are learned individuals who are willing (and perhaps able with the help of others) to initiate social and institutional changes in favor of the sciences. Historically speaking, Bacon, Descartes, and Lord Derby were examples of such individuals. The *Cultural Project*, which I note as C, regards the activity intended to initiate behavioral and moral change throughout the state. The *Institutional Project*, which I note as I, regards the activity intended to either establish or reinforce institutions for the development of the sciences. I intend to show in this model that these two projects, once both are active, have a steady, reciprocal relationship $C \rightleftharpoons I$, where each of the projects actively aid in the development of the other.

C has two primary components, and precedes I within the model. The first and most important component is the convention of teaching *scientific education on a liberal arts basis,* that is, education beginning with the careful study of ethics, history, and human psychology, among other subjects. This allows the full extent of the technological and civic forms of utility to be realized on a societal level. Teaching the sciences on a liberal arts basis has the dual function of motivating students to be *interested* in the sciences through direct engagement and historical understanding[6], and improving the civic character of the society by encourag-

[6]See Franklin 334–341.

ing the use of reason and critical analysis within and without the classroom setting[7]. This system of education therefore primes its students to become agents of scientific advancement, while also inspiring a moral, *civic* character through the study of practical philosophy from an early age. It is important to note here, however, that this liberal arts foundation will be entirely ineffective in its civic function if it is treated solely (or even predominantly) as a means for motivating the study of the sciences; it is not hard to imagine that students will gloss over important lessons which do not have an obvious application to their desired career interests, so the promotion of the intrinsic value of the liberal arts is crucial. It therefore suffices to say here that the liberal arts within this model ought to be studied for their own sake; their instrumental utility in inspiring scientific interest is no more important than this. Such systems of education are first suggested, and later instituted, through the persuasive efforts of those early learned individuals.

The second component of C may occur simultaneously to the first: scientific knowledge becomes accessible to the general public. While the first component is largely directed towards the youth, increased accessibility of scientific knowledge (including ongoing research) fosters engagement from the adult public. Bacon[8] and Descartes[9] were early proponents of this; the open publication of science results through public media meant the disruption of outdated knowledge being accepted merely by virtue of its authority. However, there are other virtues of such a process: for one, the diffusion of scientific knowledge has the potential to improve individual lives (developments in hygiene, fostering discussion) and industrial activity (developments in technology and production)[10], and it also helps to mitigate public mistrust towards the sciences through direct engagement. Additionally, this second component of C is capable early on of interaction with the first component; the publication of science results, whether they are geared for the public or for scientists internally, fosters

[7]See Smith 788.

[8]See Bacon and Sargent 119–120.

[9]See Descartes 35.

[10]See Franklin 295–297 for a similar justification in establishing the American Philosophical Society. See Bentham and Mill 105–106 for the ethical value of hygiene and especially medical science.

scientific conversation and therefore encourages scientific activity. This component is important in maintaining engagement with the sciences, but the first component is ultimately more important in gaining the public's trust; public habits are formed most directly and profoundly in childhood.

Eventually, the students of such a system of education outlined in the first component will have grown up. Understanding that all other youths within the state have been similarly educated to understand (1) the philosophical implications of the sciences, and (2) the nature of scientific work, mistrust towards the sciences has diminished. Ideally, then, the public eventually shares a general interest in the sciences[11], and depending on their wealth and degree of interest, a willingness to *invest* in them either directly or by appealing to their government to do so. This first requires either of two conditions: either institutions of learning and research for advanced students or adults (such as the Royal Society or the NSF) already exist, or if they do not exist, the investment will instead take the form of *establishment*. On the one hand, this can directly occur through the advocacy for science on account of its utility *in theory*; these considerations ultimately led to the founding of the Royal Society and the American Philosophical Society. On the other hand, the coincidence of existing scientific development and political pressure may lead politicians or individuals of social importance to recognize the utility of science, directing them to establish institutions dedicated to its advancement. For example, public figures may directly recognize the impact of scientific development on an important event, such as war or natural disaster, as was the case in the founding of the NSF and NASA. This process marks the stage where $C \to I$, that is, where cultural activity directly induces the development of institutional activity.

The first stage of I results from the establishment or improvement of the scientific research institutions. These institutions will be funded by either private or public agents. However, as a product of the cultural upbringing and education of these agents, the limitations placed

[11]This rests on the assumption that the education system suggested is effective enough to outright *eliminate* the unpopularity of the sciences, and to render the necessity of external funding as granted. The means for making the system sufficiently effective are not explored in this work. See Smith 760 ff. for one perspective.

on scientific research by these sources (as mentioned in 3.IV above) will be exclusively moral in character. This is because the cultural stability generated through C will mitigate the short-term negative effects of public funding, and the interests of private funding sources will, ideally, not drive them to bias scientific results due to the moral considerations against doing so. Instead, their understanding of the forms of utility emergent from science will direct scientific activity. Scientific institutions can simultaneously encourage ongoing public engagement with the sciences by encouraging public feedback. This allows the public to ensure that the moral character of scientific advancement is being maintained by the institutions and the researchers they fund.

The final stage of I involves the publication of institutionally-sponsored research[12], and the production of technology resulting from this research. This is chiefly the manner in which scientific researchers "pay back" funding agents for their endowment. At this point, the model has converged back to the second component of C, and the publicized research (alongside the resulting technological advancement) is valuable here for the same reasons. Therefore the structure of the model, once the institutions have been established, contains many elements whose interactions are *reciprocal*, simultaneously involving activity in C and I; the activity in each plays a role in the maintenance and development of the other. Hence there is a stability that emerges, i.e.: $C \rightleftharpoons I$, in which the main components that led to the establishment of the model, namely education, publication of scientific media, development of technology, and the public investment in scientific institutions, simultaneously occur and are conducive towards the same end of an ethical, scientifically-active society.

5 Conclusion

There are obstacles to obtaining public support for the sciences. Many of these stem from either the complicated nature of science, or misunderstandings about this nature. Writers

[12]See Bacon and Sargent 267–268 for how such publication can be part of an institutional system, but compare to the earlier 119-120.

have historically tried to convey to their audiences the social value of the sciences, and to establish the responsibilities of scientists in turn. Nevertheless, there are good historical reasons why the public may hold a negative opinion of scientific study, whether from the perception of science as clandestine and potentially harmful, or from its perceived behavioral effects on those who study it. Application of the writers' theories can mitigate many of these negative perceptions while positively impacting the sciences themselves. I have attempted to illustrate this by way of a model, highlighting that the various forms of utility which scientific work produces can be fully realized through scientific education on a liberal arts basis. This has the effect of not only enriching scientific activity, but also social and political activity, by fostering engagement between politicians, civilians, and scientists - for the discovery of truth and the benefit of people.

A Original Research Prospectus - 9/21/2025

I aim to study the philosophical justifications historically presented in favor of government funding for the sciences, particularly within the scope of European and American politics. To flesh out this objective, I have framed questions to address various aspects of the topic that I am interested in. I will first present these questions, then provide a possible trajectory for my research guided by these questions. Finally, I will summarize the steps that I propose to take as part of this research endeavor. This set of questions is motivated by prior research into the topic's history:

- How has social utility been used to justify arguments for and against funding basic scientific research? Has the definition of "utility" put forward by thinkers changed with time and place?
- What makes the sciences useful, according to those who regard them as such? What makes them not useful to those who don't?
- What other justifications have historically been offered in favor of scientific research funding, and how did these affect scientific productivity? Are these justifications part of a greater political or moral framework?

The first two questions can be addressed by studying the rhetoric of advocates for research funding. Direct examples of such advocates can be found concerning the Royal Society of London. In 1850, the Society took on the responsibility of funding scientific research on the notion that it is the "duty and interest" of civil society ("Government Aid to Scientific Research"). Calls to increase this endowment inspired more rhetoric; one speaker in 1875 appealed to patriotic sentiments in promoting public and government science funding("Lord Derby on the Endowment of Scientific Research"). Within American political history, studying the contents of the Acts establishing the NSF (U.S. Congress, National Science Foundation Act of 1950) and NASA (National Aeronautics and Space Act of 1958) (primarily focusing on how they manifest or qualify their mission statements) will elucidate the time evolution of how social utility was used as a justification for science funding in Western society.

The third question can be addressed first by studying the French Academy of the Sciences. Historians claim that shifts in perception of science's societal role were correlated to the Academy's productivity, particularly citing treatment of the sciences as a means rather than an end and pressure from economic interest groups as detrimental (Shinn). Further, the American sources above perhaps call into question whether social utility was in fact the whole/only justification given for science funding. A study of whether this is the case, based on the Acts and on related commentaries, will help characterize the relationship between politics and science in American society.

In summary, the following steps might be taken to comprise this research project: first, historical examples of government funding for scientific research in Western society will be collected and then categorized according to the authority responsible for their funding. The interests of these authorities will be compared to the contemporary public interests as spoken by advocates. The interests of the various authorities will be compared, and the effects

of their interests on the funding agent will be explored. This will include a study of the scientific productivity coincident with the authorities' interests, and whether they exhibited growth or decline. This will perhaps supply a more complete perspective on the effects of politics and ethics on scientific activity.

B Annotated (Partial) Bibliography

- Francis Bacon - The Advancement of Learning (1605)
  - *Sections read: Dedication of Part I, Part I, addressing 3 objections to the advancement of learning, Dedication of Part II, Part II chapter summaries from* De Augmentis.
  - Dedicates the first part to addressing objections to the advancement of learning – namely (1) that it should be limited as learning leads to atheism, (2) that learning has a softening effect on the population, and (3, largely) that learning has a corrupting force on one's manners.
    - He addresses (1) by distinguishing the "proud" knowledge – which swells and might lead to atheism – from knowledge of nature and universality which does not. The correcting force against this is *charity*.
    - He addresses (2) by appealing to history, on the claim that excellence and political stability have often coincided with high learning.
    - He addresses (3) by outright refutation – essentially claiming that men derive a certain "weight of duty" and selflessness from understanding their own frailty as men, and the dignity of their vocation. Also explores certain "vanities" regarding learning which may be the source of this claim, and these act as a certain constraint on science, such that it aims "to the use and benefit of man."
  - Dedicates the second part to explaining the necessity of "the public designation and expense" to the advancement of the sciences by exhaustively exploring the various divisions of science. In his dedication he outlines three objects to focus on for this end: development of the places of learning, the books of learning, and the learned persons. Here develops the notion that funding for quietness of life and the avoidance of ordinary troubles is necessary to the commitment of intellectual workers.

- Francis Bacon – Novum Organum (1620)
  - *Sections read: Preface, Aphorisms 1-68, 92-114* (These were particularly important for an understanding of his philosophy as described in the editor's introduction)
  - Dedicates the work to forming a system of establishing "progressive steps of certainty" in the sciences through experience. This is founded on an *empirical* basis: not on reason or logic, but on experiments to build an understanding of nature from particulars *upward* to greater axioms.
    - This is most explicitly formulated in his description of the *scientific method* in Aphorisms 101-107.
    - These experiments (so-called experiments of *light*), according to Bacon, must interestingly serve *no* instrumental utility, but must be purely dedicated to

discovering "causes and [lesser] axioms". (To my understanding this is an early formulation of *basic research* as a form of science. It is this kind of science that I want to base my project on. Once again note that Bacon's view of the sciences do include sciences outside of the natural sciences, such as political science.)

  - Dedicates a section to exploring the limits on knowledge as embodied in four kinds of Idols: Idols of the Tribe (Humans' bias to seek patterns where there may not necessarily be a pattern), Idols of the Cave (Limitations on an individual's experience, apart from others), Idols of the Marketplace (Limitations on words' and explanations' ability to describe phenomena, particularly as spoken to common people), and Idols of the Theater (The systems formulated by thinkers which may (or may not) be internally consistent but do not fully describe nature). Explains that an empirical basis for learning mitigates the influence of these Idols.

- Francis Bacon – New Atlantis (1627)

  - A utopian novel describing the society of Bensalem, which has an institution known as Solomon's House. This institution has the ultimate end of "knowledge of Causes and secret motions of things; and the enlarging of the bounds of the Human Empire, to the effecting of all things possible". This is done through the creation of certain preparations or instruments (laboratories) which study the various kinds of knowledge, some examples being optics, acoustics, and engineering. Employees performs specific functions within the institution which, in general, are a formulation of the scientific method as described in the Novum Organum, leading up to the publication of scientific knowledge in a generally-accessible fashion.

- Life and Professional Career [of Francis Bacon] – Rose-Mary Sargent (1999).

  - Bacon began his career in law; studied at Gray's Inn. At the same time as his law studies, he began political activity which would persist until late in life. He would not gain political or financial success until the beginning of the reign of King James I, under whom he gained the title Viscount St. Albans. During this time he was found guilty of taking gifts whilst court cases were pending, and as a result was removed from office. His retirement from politics opened the way for him to write the vast majority of his philosophy (including all of the works that I have read of his). The founding of the Royal Society is in direct consequence of his philosophical efforts and, despite much disagreement in the decades immediately following his death, his epistemology and scientific method would greatly influence the thought of the Enlightenment shortly after.

    * Also inspired the founding of the Royal Institution by Benjamin Thompson in 1799, a similar institution to the Royal Society.

- Second Royal Charter for the Royal Society of London for Improving Natural Knowledge (1663, trans. 2012)

  - This charter by Charles II established the various rights and responsibilities of the Royal Society, alongside its structure. Its objective is to promote experimental sciences for the glory of God and the advantage of man (compare Bacon's perspective on the hard sciences). Its structure, so described, is comprised of a president, a council of 20-24 persons (some of which carry specific offices), and various Fellows. The society was granted permissions to purchase land, appoint typographers, correspond with foreigners, and to build facilities for "the arranging of their affairs". Compare many of these permissions with *Solomon's House* in the *New Atlantis*. The first secretary, Henry Oldenburg, would publish the first *Philosophical Transactions of the Royal Society* – an expression of the Baconian emphasis on the publicity of science.

- Jean-Jacques Rousseau – Discourse on the Sciences and the Arts (1750)
  - This discourse won the prize for the Academy of Dijon in 1850 concerning the question of whether "the restoration of the sciences and the arts contributed to the purification of mores." Rousseau takes the negative of this question, stating that the development of scientific study, because of its apparently requisite luxury and time-commitment, takes the focus of education from that of *duties* and an understanding of the various virtues ("magnanimity, fair-mindedness, temperance, humanity, courage). This is only amplified by the development of technologies which make the sciences more accessible, primarily the printing press. Ultimately he concludes by praising the various institutions which require of their members both the cultivation of mores and scientific development; as society stands to him, *the division of power and the sciences* is the ultimate cause for the baseness of the sciences. If, however, they are united, it seems as though many of his points of disagreement will dissolve.

- Benjamin Franklin – Rules for a Club Formerly Established in Philadelphia (1732)
  - This paper outlined the various topics relevant to the Junto, and questions to be asked the members pertaining to them. Questions 3-15 showcase the practical/pragmatic utility of knowledge. Knowledge is instrumentally good insofar as it is productive of "happy effects" and generates awareness of those actions which produce "unhappy effects".

- Benjamin Franklin – A Proposal for Promoting Useful Knowledge Among the British Plantations in America (1743)
  - This outlines the scope of the American Philosophical Society: to facilitate the transfer of useful knowledge across the colonies. The members of this society were to meet once a month in Philadelphia – approximately the geographic center of the colonies – to relay their own experiments, read communications from the other members, and to disburse copies of these communications. Affirms also consistent communication with the *Royal Society* and the *Dublin Society*.
    * The Dublin Society is another learned society with early interests in agriculture and science.

- Benjamin Franklin – Proposals Relating to the Education of Youth in Pennsylvania (1749)
  - Proposes the establishment of an academy by means of charter. This is accompanied by a comprehensive scheme of educational method with sources from Milton, Locke, and others. Focuses on the study of what is most useful and what is *most ornamental*, ultimately regarding the various professions. In other words, this is an early call for technical education. These topics are to be motivated first by studying *history* – students will later study the formal principles which are characterized through historical study. As a final point, moral study is not to be neglected; see final paragraph.
    * Compare to the Baconian project as framed by Solomon's House.
    * Compare to Rousseau's point about the relationship between mores and scientific study.
- John Stuart Mill – Utilitarianism (1861)
  - Being a definitive work of the utilitarian doctrine, Mill explores the social utility of scientific progress in this work – particularly citing its potential to reduce disease. This makes the promotion of science directly conducive to the first principle of the utilitarian doctrine, or the "greatest happiness principle".

For all of the entries below, the notes used in this research project were strictly written on paper, and therefore were not annotated in the same manner as the works above.

- Adam Smith – The Wealth of Nations, Book V Chapter I (1776)
- Lord Derby on the Endowment of Scientific Research – Nature (1875)
- Government Aid to Scientific Research – Nature (1876)
- National Science Foundation Act of 1950
- National Aeronautics and Space Administration Act of 1958
- Terry Shinn - The French Science Faculty System, 1808-1914: Institutional Change and Research Potential in Mathematics and the Physical Sciences (1979)

The following works were considered, but ultimately dropped due to advice or time constraints.

- Jean-Jacques Rousseau – Discourse on the Origin of Inequality
- David Hume – Of the Rise and Progress of the Arts and Sciences (1758)
- John Dewey – The Public and its Problems (1927)

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